

Train Booking System

Introduction:

The Train Booking System is a database-driven application that facilitates the booking of train trips for passengers. It allows users to sign up, update their details, add and update trains and trips, view available seats based on specific criteria, and perform operations such as booking and canceling trips. This summary report provides an overview of the system's conceptual and physical data models and describes the implementation of the system using C# and a Microsoft SQL Server database.

Entity-Relationship Diagrams (ERD):

The Train Booking System utilizes an Entity-Relationship (ER) model to represent the relationships between various entities in the system. There are two types of ER diagrams: conceptual and physical.

2.1 Conceptual ERD:

The conceptual ERD represents the high-level view of the system, focusing on the entities and their relationships. In the Train Booking System, the main entities are:

Passengers: Represents the users of the system who can book train trips.

Admins: Represents the administrators who have additional privileges in managing trains and trips.

Trains: Represents the trains available for booking, including the number of normal and VIP seats.

Trips: Represents individual trips with specific details such as trip number, time, source, destination, and available seats.

Tickets: Represents the booked tickets, linking trips and passengers.

The relationships between these entities include one-to-many relationships (e.g., Passengers to Tickets) and many-to-one relationships (e.g., Trips to Trains). The ERD helps in understanding the structure of the system and the relationships between different entities.

2.2 Physical ERD:

The physical ERD represents the actual implementation of the database, including tables, columns, primary and foreign key constraints, and other relevant details. The physical ERD for the Train Booking System includes tables such as Trips, Trains, Passengers, Admins, and Tickets, along with their respective columns and relationships.

Database Implementation:

The Train Booking System uses a Microsoft SQL Server database to store and manage the system's data. The database consists of several tables, each representing an entity in the system. Here is a brief description of the main tables:

Trips: Stores information about individual trips, including trip ID, trip number, time, source, destination, date, and available seats.

Trains: Represents the trains available for booking, including the train number, associated trip ID, and the number of normal and VIP seats.

Passengers: Stores passenger details such as passenger ID, first name, last name, password, and email.

Admins: Stores administrator details, including admin ID, first name, last name, password, and email.

Tickets: Connects trips and passengers, representing the booked tickets.

The tables are linked through primary and foreign key constraints, ensuring data integrity and maintaining the relationships between entities.

C# Implementation:

The Train Booking System is implemented using C#, which provides the necessary functionality to interact with the database and perform various operations. The C# code includes methods to sign up new users, update user details, add and update trains and trips, retrieve available seats based on criteria, and perform booking and canceling operations.

The code utilizes the ADO.NET framework to establish a connection with the SQL Server database, execute SQL queries and commands, and retrieve data into DataTables or update the database with user input.

To interact with the database, the code includes SQL statements such as INSERT, UPDATE, DELETE, and SELECT, along with parameterized queries to ensure data security and prevent SQL injection attacks.

Conclusion:

Overall, the C# implementation provides a user-friendly interface and seamless integration with the database, allowing users to interact with the Train Booking System efficiently.